

Early Peripapillary and Macular Microvascular Changes Following Ruthenium-106 Plaque Brachytherapy For Uveal Melanomas



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- **OBJECTIVE:** To evaluate early peripapillary and macular microvascular changes following ruthenium-106 (Ru-106) episcleral plaque brachytherapy for uveal melanoma using optical coherence tomography angiography (OCTA).
- **DESIGN:** Prospective case series.
- **SUBJECTS:** Twenty-four eyes of 24 patients with extramacular uveal melanoma treated with Ru-106 plaque brachytherapy at Cairo University Ocular Oncology Service.
- **METHODS:** All patients underwent ophthalmic evaluation and OCTA imaging at baseline, 3 months, and 6 months post-treatment.
- **MAIN OUTCOME MEASURES:** Quantitative OCTA parameters included vessel density (VD) in the superficial (SCP), deep (DCP), and radial peripapillary capillary (RPC) plexuses, as well as foveal avascular zone (FAZ) area. Radiation doses to the tumor apex, fovea, and optic nerve were also recorded.
- **RESULTS:** Mean best-corrected visual acuity (BCVA) declined from 0.44 ± 0.30 logMAR at baseline to 0.78 ± 0.72 at 6 months ($p = .028$). Tumor height decreased significantly ($p < .001$). No clinical signs of radiation maculopathy or optic neuropathy were detected at 6 months. RPC density showed an early significant reduction ($49.06 \pm 2.77\%$ to $44.95 \pm 4.65\%$ at 3 months, $p < .001$), with no further decline at 6 months; this loss was greater in tumors < 2 disc diameters from the optic disc. SCP showed localized reductions ($46.23 \pm 3.73\%$ at baseline, $43.96 \pm 5.15\%$ at 3 months and $44.64 \pm 4.26\%$ at 6 months, $p = .083$). By contrast, DCP density declined progressively and diffusely ($45.17 \pm 4.86\%$ to $44.90 \pm 4.87\%$ at 3 months and $42.23 \pm 3.68\%$ at 6 months, $p = .017$), accompanied by significant FAZ enlargement (0.25 ± 0.12 to 0.34 ± 0.11 mm²,

$p < .001$). Higher optic nerve radiation doses correlated with RPC loss ($r = 0.492$, $p = .014$) and poorer BCVA at 3 months ($r = -0.454$, $p = .026$) and 6 months ($r = -0.412$, $p = .045$), whereas macular doses showed no significant correlation. Tumor location influenced vascular response, with post-equatorial tumors showing greater DCP compromise, and pre-equatorial tumors exhibiting more SCP reduction. Systemic comorbidities did not significantly alter outcomes.

- **CONCLUSIONS:** Ru-106 brachytherapy induces early, subclinical microvascular injury detected by OCTA. RPC changes are early and strongly correlate with optic nerve radiation dose and visual decline, SCP changes are localized, while DCP demonstrates progressive and generalized loss. Tumor location influenced injury patterns, reflecting differential radiation exposure. OCTA may serve as a noninvasive biomarker for early detection and risk stratification of radiation-induced microvascular injury. (Am J Ophthalmol 2026;284: 133–143. © 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.)

INTRODUCTION

UVEAL MELANOMA (UM) IS THE MOST COMMON primary intraocular malignancy in adults, with an incidence of about 4.3 per million.¹ Historically managed with enucleation, eye-sparing treatment with episcleral plaque brachytherapy gained wide acceptance after the Collaborative Ocular Melanoma Study (COMS), achieving high rates of local tumor control and globe salvage.² However, visual outcomes remain limited; in COMS, nearly half of patients had severe visual impairment within 3 years of treatment.³ Post-brachytherapy vision loss may result from potentially reversible causes such as radiation cataract, but more often from irreversible vascular complications including radiation retinopathy (RR), radiation maculopathy (RM), and radiation optic neuropathy (RON). These complications stem from radiation-

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